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Twenty-Five Years of Hub Location Research

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Last year was the 25th anniversary of two seminal transportation hub location publications, which appeared in 1986 in *Transportation Science* and *Geographical Analysis*. Though there are related hub location and network design articles that predate these works, the 1986 publications provided a key impetus for the growth of hub location as a distinct research area. This paper is not intended as a comprehensive review of hub location literature; rather, our goal is to reflect on the origins of hub location research, especially in transportation, and provide some commentary on the present status of the field. We provide insight into early motivations for analyzing hub location problems and describe linkages to problems in location analysis and network design. We also highlight some of the most recent research, discuss some shortcomings of hub location research and suggest promising directions for future effort.

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1. Introduction

The year 2011 marked the 25th anniversary of the beginnings of transportation hub location as an important sub-field of location science. Hub location has been a fertile area for interdisciplinary researchers from operations research, transportation, geography, network design, telecommunications, regional science, economics, etc, and the field is now well known with excellent overview treatment in both introductory and advanced texts (Taaffe, Gauthier, and O'Kelly 1996; Daskin 1995; Drezner and Hamacher 2002). This paper is not intended as a comprehensive survey or review of hub location research; for that, see Alumur and Kara (2008); Campbell, Ernst, and Krishnamoorthy (2002); and Klincewicz (1998). Our key contributions are to provide some reflections on the origins of transportation-oriented hub location research, brief commentary on the present status of hub location research, and suggestions for some promising directions for the next 25 years of hub location research. Along the way we also allude to many variants of the problem and remark on the number of allied problems that both provide useful methods and stimulate conceptual network design. Although our primary interests are in hub location for transportation networks, there is a large body of related research on network design and on location in telecommunications networks, and we highlight some of the linkages between these areas in §2.

Hub location problems involve the location of hub facilities through which flows (e.g., of passengers or freight) are to be routed from origins to destinations (e.g., cities). Origin-destination (OD) trips from node i to node j are associated with OD pairs, which may be connected with *direct* trips (not via hubs) or via paths that visit hubs (see Figure 1). This diagram shows three possible paths from origin i to destination j: path 1 is a direct trip (the direct point-to-point arrow from i to j); path 2 is a one stop trip via hub h; and path 3 is a two stop trip via hubs h_1 and h_2. Note that the traffic (e.g., passengers) transiting through h, or through h_1 and h_2, has switching options at the hubs where it can be joined by traffic (e.g., other passengers) arriving from other origins (the in-arrows) and can depart to other destinations (the out-arrows).

Hub location problems require the location of one or more hub facilities, which can provide two major functions: (1) a switching, sorting, or connecting (SSC) function that allows flows to be redirected at a hub node (enter via one link and depart via another) and (2) a consolidation/breakbulk (CB) function that allows flows to be aggregated and disaggregated. The SSC function allows redirection of flows based on the destination and thereby allows many origins and destinations to be connected with many fewer links than would be required with direct connections. The CB function allows reduced costs for consolidated flows by exploiting economies of scale, which are especially

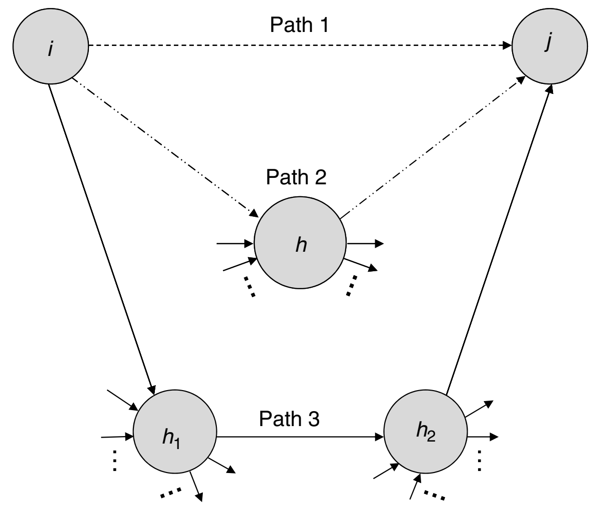

Figure 1 **Alternative Paths Between Origin *i* and Destination *j***

strong in transportation; they serve therefore to concentrate or de-concentrate traffic. Note that the CB function of a hub is useful even on links not between hubs because routing OD flows via hubs allows for concentration of flows on the links between origins or destinations and hubs. We should point out that the term "hub" is often used in a general sense in several different ways: it may be a facility through which large amounts of traffic connect (but not including any originating or terminating traffic) or a facility through which large amounts of traffic pass, including all originating and terminating traffic (see, for example, Chou 1990; Jaillet, Song, and Yu 1996; Lium, Crainic, and Wallace 2009). This *a posteriori* definition of a hub based on traffic flows differs from the common definition in hub location research where hubs are distinct facilities to be located that can provide the SSC and CB functions. To be clear, this paper refers to hubs primarily in terms of their role as SSC and CB facilities. Also note that in some cases functions other than SSC and CB may be performed at hubs, including some processing that might change the nature of transportation and splitting/combining as for packet switching in telecom networks.

Hub networks are very common in large-scale transportation and telecommunications systems because they can help to reduce costs of providing the service, increase service frequencies, and hedge against uncertainty in demand (Lium, Crainic, and Wallace 2009). Note that for many transportation hub networks, the transportation links are public infrastructure (e.g., road, air, sea), so the establishment of the transport links is often not a key concern in the hub location model. The main focus is the cost of travel on these links—and opportunities to exploit transportation economies of scale. However, the public or private nature of the transportation infrastructure depends on the mode (e.g., rail versus road in the United States) and government transportation policies (e.g., private versus national carriers in some countries). There can be many similarities in telecom hub location models, although in telecom networks there are no vehicles and the predominant costs are often associated with establishing the physical links (e.g., fiber optic lines). Usage of the links may incur a (generally low) cost per unit of flow, though pricing for usage of the links often reflects economies of scale to cover the costs of establishing and operating the links and hubs.

The remainder of the paper is organized as follows. Section 2 provides details of hub location problems and discusses the linkages between relevant problems in hub location and network design. Section 3 describes the early development of hub location as a research area, including important practical motivations. Section 4 highlights some of the most recent hub location research and offers some commentary on some research directions. Section 5 describes some promising research directions, and §6 includes our conclusions.

2. Hub Location Problems

This section starts with a broad definition of hub location problems and discusses the relationship between hub location problems and network design problems. We describe the components of hub networks, briefly discuss transportation and telecom hub networks, and define fundamental hub location problems that have been the focus of much research for the past quarter-century.

2.1. Distinguishing Features for Hub Location Problems

We begin with a very broad perspective on hub location, for which the key distinguishing features are the following:

1. Demand is associated with flows between OD pairs (not with individual points).
2. Flows are allowed to go through hub facilities.
3. Hubs are facilities to be located.
4. There is a benefit of routing flows via hubs (or a requirement to route flows via hubs).
5. There is an objective that depends on the locations of hub facilities and the routing of the flows.

More formally, we can consider a complete graph $G = (V, E)$ with node set $V = \{1, 2, \ldots, N\}$, where nodes correspond to origins, destinations, and potential hub locations. One could easily have a separate set of possible hub locations rather than using V. Each arc (i, j) is of length d_{ij}, and W_{ij} is the flow (e.g., amount of freight, passengers, or data) to be transported from i to j. Throughout this paper, we assume the lengths d_{ij} are Euclidean and thus satisfy the triangle inequality. In this general problem statement, flows are allowed, though not required, to go through hubs, so the incentive to use the hubs is provided by item 4. This might be because of a reduction in the number of links required to connect the network, a reduction in transport cost from economies of scale (whether between hubs or not), or something else reflected in the objective or constraints of the problem (e.g., all flows must visit at least one hub or the number of links [e.g., aircraft] is limited). Note that in a profit maximizing model, the result may be that some demands are not served.

With the above generic description of hub location problems, the main decisions are the location of hub facilities and the routing of flows to serve the demand. The routing of flows implicitly defines a hub network consisting of (i) origin and destination nodes, (ii) hub nodes, (iii) *access arcs* that connect non-hub origin and destination nodes to hub nodes, (iv) *hub arcs* that connect two hub nodes, and possibly (v) arcs that directly connect two non-hub origin and destination nodes. These main decisions are very much in the tradition of location theory where two of the distinctive decisions are location of facilities and allocation of demand points to facilities (reflecting the routing of flow between the facilities and demand points). Depending on the particular hub location problem variant, there may also be other decisions, such as selecting the type or capacity of the hub when multiple options are available.

Just as for most facility location problems, the generic hub location problem can be viewed as a network design problem with location. See Contreras and Fernández (2012) for a review of combined network design and location problems. Items 1–5 distinguish hub location problems from many related problems, including network flow problems without location and location problems without OD specific flows (e.g., the transshipment problem). However, many multicommodity network design problems include OD specific demands either explicitly or implicitly. For example, a network that requires n nodes to be connected can be viewed as one designed to serve a demand for travel or communication between all $n(n-1)$ node pairs (i.e., $W_{ij} = 1$ for all i, j). Although many such network design problems do not have a facility location component (e.g., the minimum spanning tree problem), those with a facility location component could be viewed as hub location problems. For example, the Steiner Tree Problem could be viewed as hub location problem where the Steiner points are hubs and the objective is to minimize the length of the network to connect all flows, though this is traditionally and quite logically viewed as a network design problem and not a location problem. This illustrates the broad overlap between hub location and network design problems, as noted in O'Kelly (1986b). From this broad perspective, Miehle (1958, p. 232), which provides physical and mathematical models to locate interconnection points described as "communication centers, road junctions, or distribution centers" that connect fixed points with edges of minimum total length, could be viewed as one of the earliest hub location papers.

Two large classes of network design problems that can be viewed from a hub location perspective are (1) multicommodity network design problems, where commodities are associated with flows between OD pairs and the network design includes locating intermediate nodes, and (2) network design problems that require connectivity via intermediate nodes to be located. This includes many two-level network design problems with location, where the upper level is the hub facilities and hub arcs and the lower level consists of the demand points and access arcs. In telecommunication networks, the upper level is often termed the backbone level and the lower level is the tributary or access network.

The dual perspectives of facility location and network design apply to many problems although particular restrictions and assumptions may emphasize one domain over another. For example, locational aspects of a problem may be diminished by assuming or restricting the set of facility locations and network design aspects can be diminished by assuming a particular topology. Whether a problem is viewed as network design with location or location with network design may be as much a historical artifact as a conscious decision; both areas of research contribute valuable perspectives to solving hub location problems. Although hub location problems can be broadly defined as above, in the last 25 years the majority of research that has used the label "hub location" has addressed a restricted problem with a maximum of two hub stops. Consequently, many hub location models have been tailored for these problems, and alternative models may be needed for more general hub location and network design problems.

2.2. Applications

We are concerned with location of physical facilities, so the main application areas of hub location have been in transportation and telecommunications. (There may well be hub location problems in more abstract settings, such as social networks, where distance is not a measure of geographical displacement.) Transportation hub location problems involve locating hub facilities to handle flows of freight or passengers. Transportation is accomplished by the movement of vehicles carrying the freight or passengers on infrastructure such as roads or railways or through the air or water, for which there is an associated cost and/or travel time that depends on the distance traveled. Telecommunications hub location problems involve locating hub facilities (i.e., hardware such as switches, routers, and concentrators) to provide communication among a set of nodes (origins and destinations), possibly with demand specified as OD flows W_{ij}. Telecommunications hub networks provide for the movement of electronic data using physical links (e.g., cables, fiber links) or though the air (e.g., microwaves), rather than the movement of tangible objects in vehicles, and there may or may not be significant costs that depend on the distance the message travels. Transportation and telecom hub networks often have a similar appearance, and abstract models for problems in both domains can be very similar. However, the operations and consequently the relevant costs, service measures, and constraints are often quite different because of the differing natures of physical objects (freight or passengers) and electronic signals. Also note that routing protocols for transportation and telecom may differ, such as packet switching in large telecom networks; in contrast, individual traffic units in transportation networks (e.g., passengers or shipments) are usually not divisible.

Some hub location research employs idealized models with only a generic reference to possible applications in transportation or telecom, whereas other research has developed application specific models with associated constraints and objectives. Transportation hub location research has generally focused on transport costs and travel times to serve a given demand W_{ij}. The considerably more difficult equilibrium adjustment of interactions to network structure is largely unexamined, although endogenous flows were considered in O'Kelly (1986a). Telecom hub location research has focused much attention on fixed costs for establishing a network (as with fixed charge network design problems) and on delays and congestion in the network (which often leads to a probabilistic perspective and consideration of queuing issues). However, these issues can arise in either domain depending on the particular application.

2.3. Fundamental Hub Location Problems

The generic definition of hub location problems in items 1–5 encompasses a wide variety of different problems, many of which have not traditionally been viewed as hub location problems (nor should they be). From the facility location research perspective, hub location problems were recognized early on as being related to well-established location problems, yet posing new challenges. Campbell (1994b) defined a fundamental set of hub location problems, analogous to the fundamental regular facility location problems that have served as a starting point for much subsequent research. Although the more general problem as in items 1–5 is consistent with early surveys of hub location research by Campbell (1994a) and O'Kelly and Miller (1994), the fundamental hub location problems defined in Campbell (1994b) include two important features:

6. Paths between OD pairs visit at most two hubs.
7. Direct OD flows are not allowed.

Fundamental hub location problems also generally assume that each non-hub either is linked to a single hub, termed single allocation, or can be linked to more than one hub, termed multiple allocation. Both situations occur in practice: for example, passenger airline networks typically have multiple allocation because there are flights from some non-hub cities to several or all of an airline's hubs, whereas less-than-truckload (LTL) trucking networks may have each non-hub node (i.e., end-of-line terminal) assigned to a single breakbulk terminal (i.e., hub). Similarly, some telecom networks employ single allocation with star access networks to reduce the cost to construct the network, and others allow or require multiple allocation, as for example to increase reliability and/or provide backups (e.g., Carello et al. 2004). Also note that capacities on flows at hubs may force multiple allocation, as with many facility location problems.

Item 6 is often reflected in hub location research as a topological requirement that the hubs are fully interconnected by hub arcs; however, that is more properly a result of some basic modeling features analogous to those in regular facility location problems, such as the p-median problem. The p-median is a fundamental discrete facility location problem, not just because of its long history, but because it relies on a minimum of assumptions: Given a set of demands at points (nodes), locate p facilities at candidate sites to minimize the total travel distance (cost or time) to serve the demands. Because the demand is viewed as being served by movement between a facility and a demand point, the total distance is the demand weighted sum of distances between a demand point and its closest facility. Thus, the property that each demand point is allocated to a single facility is a byproduct of the solution for the p-median problem.

The hub location analogue to the p-median is the p-hub median defined by Campbell (1994b): Given a set of demands (flows for OD pairs), locate p-hub facilities at candidate sites to minimize the total transportation cost to serve the demand. Because demand in transportation hub location problems is for service from an origin to a destination, the total transportation cost is the demand weighted sum of costs for serving all OD pairs. Thus, each OD pair is routed separately to minimize total transportation costs, and multiple allocation is a byproduct of the solution for the p-hub median problem. (The p-hub median with single allocation is then a variant of the p-hub median with a restriction of single allocation.) The key in this model is, then, how the cost for an OD pair is specified. When there is a single hub (e.g., at node k), then the natural definition for the transportation cost per unit of flow to serve OD pair (i, j) via the hub is $\delta(d_{ik} + d_{kj})$, where δ is the transportation cost rate (e.g., dollars per unit of demand per mile). With two or more hubs in a network, there is a possibility of movement between hubs and a simple way to model scale effects on those links is via a constant discount factor. (This basic idea was adopted independently by Goldman 1969; O'Kelly 1986a, 1987; and Campbell 1992.) Thus, the transportation cost per unit of flow for OD pair (i, j) served via hubs k and m in that order, denoted C_{ijkm}, can be modeled as $C_{ijkm} = \chi d_{ik} + \alpha d_{km} + \delta d_{mj}$, where χ, α, and δ are transportation rates for collection from the origin to hub k, transfer between hubs k and m, and distribution from hub m to the destination, respectively. Generally, it is assumed that the transportation rates are constant for each type of movement (collection, transfer, and distribution) throughout the network and the rate for transfer is less than that for collection and distribution, so $\alpha < \chi$ and $\alpha < \delta$. Further, it is often assumed that the collection and distribution rates are equal and measured in units such that $\chi = \delta = 1$, so that the interhub transportation discount is captured by $0 \le \alpha \le 1$.

With this model of transport cost discounted by α between two hubs, where distances follow the triangle inequality (as for Euclidean distance)—and absent any other restrictions or assumptions—then the transportation cost is minimized when paths between OD pairs visit at most two hubs. When there is a nonzero flow between all OD pairs, then this property implies full interconnectivity of hubs by hub arcs. From this perspective, item 6 (paths between OD pairs visit at most two hubs) is not an assumption but a consequence of the assumption that cost is discounted between any two hubs, presumably to reflect scale economies, along with the model for cost on an arc as proportional to the arc length. (Note that if the distance metric was such that the triangle inequality did not hold, then paths between OD pairs would not necessarily visit at most two hubs.) A more complex and realistic approach to transportation cost would have the cost dependent on the flows on the arcs, as for example in a concave cost multicommodity network flow model (e.g., Minoux 1989). Also, note that in some settings the interhub cost could be quite different, as in Helme and Magnanti (1989) where the interhub cost per unit of traffic is independent of the interhub distance.

The remaining issue in defining the fundamental hub location problem setting is the prohibition of direct OD shipments (item 7). The justification for item 7 is often based on removing from the demand set any flows that are large enough to be sent directly, thereby leaving only the flow that would travel via hubs. As with many properties of the hub location problem, this intuitive and relatively obvious strategy may actually be an inappropriate representation of the true problem because the opportunity to route these flows through hubs, though seemingly circuitous, can help the network achieve economies of scale. As the field has matured and the treatment of flows has become more sophisticated, the literature has moved to incorporate more realistic aspects of logistics and flow (e.g., Groothedde 2005). A major opportunity remains in linking this field more securely to topics from supply chain and inventory management, though there have been some excellent starting points on this front (e.g., Hall 1989; Groothedde, Ruijgrok, and Tavasszy 2005; Cheong, Bhatnagar, and Graves 2007; Smilowitz and Daganzo 2007).

Using items 1–7, Campbell (1994b) defined four fundamental hub location problems, analogous to the fundamental facility location problems: p-hub median, uncapacitated hub location, p-hub center, and hub covering. The p-hub median problem was originally formulated using variables that track flows on three-part paths consisting of collection (origin-hub), transfer (hub-hub), and distribution (hub-destination): X_{ijkm} = the fraction of flow from origin i to destination j on path i-k-m-j (in that order). Binary variables were used to indicate hub locations: $Y_k = 1$ if node k is a hub, and 0 otherwise. An efficient formulation based on the approach in Marín, Cánovas, and Landete (2006) is

$$\text{Minimize} \quad \sum_{i,j,k,m} W_{ij} C_{ijkm} X_{ijkm} \tag{1}$$

$$\text{Subject to} \quad \sum_{k} \sum_{m} X_{ijkm} = 1 \quad \forall\, i, j \in V, \tag{2}$$

$$\sum_{k \in V} Y_k = p, \tag{3}$$

$$X_{ijkk} + \sum_{m \ne k} (X_{ijkm} + X_{ijmk}) \le Y_k \quad \forall\, i, j, k \in V, \tag{4}$$

$$Y_k \in \{0, 1\} \quad \forall\, k \in V, \tag{5}$$

$$X_{ijkm} \ge 0 \quad \forall\, i, j, k, m \in V. \tag{6}$$

Note that this formulation allows multiple allocation for all nodes because the X_{ijkm} variable represents the OD paths. The objective is to minimize the total cost for collection, distribution, and transfer. Constraint (2) ensures that each OD flow is sent via some hub pair (possibly a single hub as in X_{ijkk}). Constraint (3) requires that exactly p hubs are selected. Constraint (4) ensures that hubs are opened for all routings of flows. Constraints (5) and (6) restrict the variables appropriately. This model includes $O(|V|^4)$ variables for all the path flows and $O(|V|^3)$ constraints.

The single allocation p-hub median problem was originally formulated by O'Kelly (1987) as a quadratic integer program using assignment variables: $Z_{ik}=1$ if location i is allocated to the hub at location k, and 0 otherwise. With this approach Z_{kk} indicates hub locations and the single allocation p-hub median problem is

$$\text{Minimize} \quad \sum_{i,j,k,m} W_{ij}(Z_{ik}c_{ik} + \alpha Z_{ik}Z_{jm}c_{km} + Z_{jm}c_{jm}) \tag{7}$$

$$\text{Subject to} \quad \sum_{k} Z_{ik} = 1 \quad \forall\, i \in V, \tag{8}$$

$$\sum_{k\in V} Z_{kk} = p, \tag{9}$$

$$Z_{ik} \le Z_{kk} \quad \forall\, i, k \in V, \tag{10}$$

$$Z_{ik} \in \{0, 1\} \quad \forall\, i, k \in V. \tag{11}$$

Constraints (8)–(11) are analogous to (2)–(6). This model includes $O(|V|^2)$ constraints and $O(|V|^2)$ variables, many fewer variables than in the multiple allocation p-hub median formulation, but the objective includes quadratic terms for the interhub transportation cost, yielding a nonconvex problem. The other fundamental hub location problems can be defined and formulated similarly (Campbell 1994b; Campbell, Ernst, and Krishnamoorthy 2002; Alumur and Kara 2008). Note that hub covering problems are less straightforward than are the analogous regular facility covering problem because coverage may be based on OD paths or on individual components (e.g., arcs) of the paths. Campbell (1994b) defined three types of coverage and corresponding hub center problems.

2.4. Challenges in Hub Location

Hub location problems are NP-hard, except in very special cases, and this difficulty stems from the inclusion of elements of both facility location problems and the quadratic assignment problem because of the interhub flows. Because both of these problems are very difficult by themselves, their combination tends to make hub location problems at least as difficult as the analogous regular (non-hub) facility location problem. The majority of hub location research has addressed network and discrete problems, and researchers have employed the full range of standard optimal and heuristic solution approaches to solve hub location problems. Some brief details on formulations and solution methodologies are provided in later sections.

The fundamental hub location models have been extended in many ways, analogous to the extensions in facility location research (e.g., with capacities, competition, reliabilities, stochasticity, etc.), but also with features from network design problems (e.g., hop constraints and restricted topologies). This blend of location and network design aspects gives rise to special challenges, as for example with capacities that may be upper and lower limits on the flows on the arcs, on the flows entering hubs from non-hubs, or on the flows into or through the hubs. A variety of topologies, in addition to complete graphs as discussed earlier, may also be reasonable, including stars, trees, and more general shapes. Some examples are discussed in §4, with more details provided in Alumur and Kara (2008) and Campbell, Ernst, and Krishnamoorthy (2002).

3. Development of Hub Location as a Research Area

The first hub location publications (as we think of the field now) appeared in 1986 in *Transportation Science* and *Geographical Analysis* (O'Kelly 1986a, b). However, earlier works such as the seminal papers in network location by Hakimi (1964, 1965), which prove node-optimality for the one-median and p-median problem, were actually motivated by hub location problems. The first two sentences of Hakimi (1964, p. 450) describe OD flows in "a communication network, such as a telephone interconnecting system … [where] all traffic flows (messages) within the network must arrive at the center S before they are processed and then sent to their proper destination." Because this paper addresses location of a single facility, the OD flows are converted to equivalent nodal demands, which then produces the one-median problem. Similarly, O'Kelly (1986a) converts the continuous planar one-hub median problem to the Weber problem. The follow-up paper by Hakimi (1965, p. 462) has a similar motivation to locate switching centers, but here movement is possible between two centers as in a hub network. However, this possibility is quickly eliminated ("It is assumed that the cost of providing the necessary links between switching centers is negligible"), and the paper then focuses on the p-median. Although Hakimi (1964, 1965) clearly had the idea of a hub in the title, the reduction to either a one-median, where interhub flows are moot, or to a multimedian, with zero cost for interhub movements (i.e., $\alpha = 0$), in our view missed the opportunity to deal with interdependent facilities.

Inspired by Hakimi (1964, 1965), Goldman (1969, p. 353) extended node optimality to what are essentially transportation hub location problems by modeling transportation with different rates for collection to a first facility where an initial processing occurs, transfer to a second facility where final processing occurs, and then distribution to the designated destination. The context for this work is the transport of materials with multistage processing at facilities, where the processing may include alterations to the "degree and nature of consolidation." Goldman also explicitly mentions postal operations as an application and notes the importance of economies of scale as "center-to-center movement is typically of a bulk and/or long haul variety, likely to have a lower unit cost" than for collection and distribution. Although Goldman (1969) proves node optimality for facilities with this type of transportation cost, he does not provide a solution algorithm or numerical examples. Interestingly, although Goldman published five subsequent papers in *Transportation Science* between 1970 and 1972, none addressed the type of hub networks in Goldman (1969), and this paper was relatively forgotten until its rediscovery and linkage to hub location by Campbell around 1990. If researchers had followed up on the hub location model in Goldman (1969), then perhaps this area of research would have developed much sooner and Goldman (1969) would occupy a similar position in hub location to that of Hakimi (1964, 1965) in facility location. The reasons that Goldman (1969) was largely overlooked may be because of the currency of nodal demand analysis in vogue in those early days or perhaps because the practical motivations were not yet strong enough.

3.1. Motivations

Important motivations for the emergence of research on hub location were developments in the 1970s and 1980s regarding deregulation of transportation, the rise of express delivery services, the development of large telecommunications networks, and advances in modeling. Each of these is described briefly in this section.

Beginning in the late 1970s, deregulation of transportation in the United States spread swiftly across all modes, encompassing air cargo in 1977 (the Air Cargo Deregulation Act), passenger airlines in 1978 (the Airline Deregulation Act), and trucking companies and railroads in 1980 (the Motor Carrier Act of 1980 and the Staggers Rail Act, respectively). In addition to removing or relaxing price controls, deregulation reduced barriers to entry, expansion, mergers, and acquisitions. This led to considerable focus by transportation firms (especially airlines and trucking companies) on larger-scale transportation service networks, including optimizing the terminal (or hub) locations because carriers could now use their networks and services to achieve competitive advantages. Hub transportation networks were deployed in short order throughout much of the passenger and cargo airline and LTL trucking industries.

Another important practical motivation for hub location research was the advent of express delivery firms, most notably Federal Express (FedEx). Air express has existed in some form since nearly the beginnings of air transport, and several national postal networks had hub-like structures (Allaz 2005). In fact, India established an overnight airmail service in 1949 using a simple one hub system to connect four major cities (Bombay, Calcutta, Madras, and New Delhi) via the centrally located city of Nagpur. Operations in this small network were nearly identical to those later used for overnight service by FedEx, with nightly departures from the four spoke cities to Nagpur (scheduled between 9:25 P.M. and 9:55 P.M.) and return trips from Nagpur arriving back at the spoke cities between 5:50 and 6:05 A.M. (Bayanwala 2011). In spite of the prevalence of postal and freight transportation networks, the advent of Federal Express in 1973 (initially serving 11 cities via a single hub in Memphis—but quickly expanding to a more successful 26-city network) marked a turning point in the recognition of the value provided by hub networks (Chan and Ponder 1979; Mason et al. 1997). From its inception, Federal Express worked for regulatory reforms to permit more efficient operations, especially the use of larger freight aircraft that was skillfully advocated by Fred Smith as part of air cargo deregulation, which also provided specific advantages designed for FedEx (Fisch 2005). Federal Express has also long relied on quantitative analysis and was using OR models to locate hubs as early as 1979 (Mason et al. 1997). Federal Express also made an important indirect early contribution to academic hub location research with an exhibit at the National Air and Space Museum in Washington, DC, and an accompanying press release (Smithsonian Institution 1983) that was very instrumental in O'Kelly viewing the hub location problem as amenable to the tools of standard location theory.

In addition to this renewed focus on transportation networks, the 1970s and 1980s also saw the creation of large computer communications and telecommunications networks. Though much of the research focused on topological design of networks for given backbone nodes (hubs), publications addressing location of the hubs date back to at least the 1970s (Boorstyn and Frank 1977; Gerla and Kleinrock 1977). This research includes both general backbone topologies (e.g., Gavish 1992) as well as restricted topologies such as trees (Kim and Tcha 1992), rings (Lee, Ro, and Tcha 1993), and complete graphs (Helme and Magnanti 1989; Chung, Myung, and Tcha 1992). For reviews of this work, see Campbell (1994a); Klincewicz (1998); and Gourdin, Labbé, and Yaman (2002). There has certainly been a wealth of research on telecom hub location since these reviews, but another comprehensive review of telecom hub location is beyond the scope of this paper.

Advances in OR models in location science also provided early motivation for hub location research and some historical perspectives on location science and locational analysis are provided by Hale and Moberg (2003); Smith, Laporte, and Harper (2009); and Eiselt and Marianov (2011). Increasing computing power also led to new algorithmic approaches and the solution of larger and more complex models. Another important area of modeling advances was in continuous approximation modeling for many-to-many logistics systems; see Daganzo (2005) for an overview. In these models, demand is treated as continuously distributed over a (planar) service region, and solutions featured the allocation of origins and destinations to multiple facilities. Models often considered tradeoffs between transportation and inventory or facility costs and were used to design optimal networks with transshipments via hubs (for example, Daganzo 1987; Hall 1984).

Within the field of geography, gravity models and their descendents, spatial interaction models, have long been a foundation of models for movement. Typically, transport geographers view the flows between places as amenable to systematic descriptive and explanatory models of movement, such as the work in Olsson (1970) and Fotheringham (1981). Geographers also had a long record of analyzing locations for manufacturing industry and warehouses for distribution, and the advent of location-allocation models opened a new vista by virtue of the ability to locate other kinds of public and private facilities. Geographers, however, tended to treat network analysis and spatial interaction as two separate topics. Clearly the field was ripe for a connection, seeing flows and network infrastructure as dual aspects of the same problem. This was initially explored, and provided some of the foundation for hubs, by work linking the location allocation model to the spatial interaction framework (one notable early example is Hodgson 1978).

3.2. Early Hub Location Research Efforts

Morton O'Kelly provided seminal publications in hub location (O'Kelly 1986a, 1986b, 1987) as an outgrowth of his research and interests in spatial interaction and location, which had its roots in training in course work at McMaster University with George Wesolowsky and Mike Webber. The original quadratic integer programming formulation for the single allocation p-hub median problem from O'Kelly (1987) used a constraint set identical to that for the p-median problem (constraints (8)–(11)), which was very familiar to geographers through the classic publication by Revelle and Swain (1970), and an objective similar to the quadratic assignment problem, which was familiar to geographers and planners as a model for resource allocation (Snickars 1978). Because of the known difficulty of nonconvex quadratic programs, this model was derived largely by graphical and numerical calculation of examples and was initially thought to possess many of the properties of the p-median; however, exploration of the solution space showed that nonclosest assignment could be optimal. Appearing about the same time were two other papers, O'Kelly and Grove (1985) and Grove and O'Kelly (1986), which were not so focused on location as on addressing congestion issues in hub networks.

In these early years, both continuous and discrete location models followed roughly parallel paths of development, in contrast to work today that tends to be specialized to the point that investigators are either planar or discrete modelers (there are notable exceptions, of course). This early work also introduced the CAB data set for 1970, those data having already been widely used by geographers as a test case for spatial interaction models (Fotheringham 1981). It remains one of the standard data sets for testing new hub location algorithms and models and the 25 node data are on OR-Library (Beasley 2011).

An important forum for early hub location research were the triennial International Symposium on Locational Decisions (ISOLDE) meetings. Early versions of O'Kelly (1986a) and O'Kelly and Miller (1991) were presented at ISOLDE III in 1984 and at ISOLDE IV in 1987, respectively. In both cases, generous and highly relevant advice was provided by the audience. At ISOLDE V in 1990, O'Kelly presented an early version of O'Kelly (1992), which examined single allocation hub models with squared Euclidean distances (a type of cluster problem), whereas Campbell introduced multiple allocation hub location problems, which arose from the more natural conceptualization of hub network design from a trucking perspective.

Another source of hub location research was the discrete demand analogue of continuous approximation models of many-to-many distribution with transshipments (Campbell 1986). Campbell's interest in these problems was sparked by discussions with west coast trucking firms seeking to understand network changes in response to mergers and acquisitions made possible with deregulation. Campbell's conception of hub location problems was with multiple allocation, as consistent with continuous approximation models. Campbell (1992) provided the first linear integer programming formulation of the multiple allocation p-hub median problem using the path flow X_{ijkm} variables. These can be viewed analogous to the binary single allocation variables X_{ab} that identify whether OD pair (or commodity) a is allocated to hub pair b, though the hubs pairs must then be linked to individual hubs. Multiple allocation hub location models provide greater flexibility in modeling traffic flows—but at the expense of a large increase in the number of variables and the associated computational burden.

As hub location research developed, the use of unique terminologies, notations, and naming conventions by different authors created a need for structure and organization in the new field. In response, 1994 saw publication of three papers that provided surveys/reviews of the new field and documented the breadth of problems and research opportunities (Campbell 1994a, b; O'Kelly and Miller 1994). These works helped provide a more unified view of hub location research and placed the hub models in the comparative context of the existing body of literature in location science. This tactic proved to be a fruitful means to generalize a broad family of models, and quite quickly a range of discrete and planar models over median and covering objectives was developed. These early efforts to bring greater coherence to hub location research also resulted in the first special journal issue on hub location in 1996 (*Location Science*, Volume 6(3)).

At about the same time as these efforts to review and organize the field, a more applied research project was underway with Australia Post (AP). In 1994, AP requested that the operations research group at Commonwealth Scientific and Industrial Research Organization (CSIRO) model their mail sorting and distribution center (SDC) network in Sydney to assess whether the continued use of a single large SDC in central Sydney was efficient. Mohan Krishnamoorthy began this work at CSIRO and quickly realized that the AP problem was a version of the newly published p-hub median (where SDCs are hubs). Although much of the literature was solving problems with 25 nodes (cities), AP was interested in solving larger problems of about 200 nodes (postcodes) and about 10 hubs (SDCs). CSIRO developed a simulated annealing heuristic solution algorithm and results were used by AP in an internal report (and later in Ernst and Krishnamoorthy 1996). Following the contract work for AP, CSIRO researchers (Krishnamoorthy and Andreas Ernst) undertook efforts to solve larger problems to optimality using the AP data set (with up to 200 nodes and 10 hubs). A key contribution by Ernst and Krishnamoorthy was the development of smaller formulations with three-subscript variables that track flows by origin on each arc: X_{ik} = the flow from origin i to hub k, Y^i_{kl} = the flow from origin i that is routed between hubs k and l, and X^i_{lj} = the flow from origin i that is routed from hub l to destination j. This new formulation produced linear integer programming formulations with only $O(|V|^3)$ variables and, with further work to identify strengthening cuts, led to the solution of larger problems.

3.3. Methodological Issues

The previous section described some of the earliest work in hub location as well as three alternative formulation approaches with path flow variables (X_{ijkm}), binary allocation variables for single allocation problems (X_{ik}), and arc flow variables (X_{ik}, Y^i_{kl}, and X^i_{lj}). Once hub location problems became widely known, many problem variants were posed and solved using a variety of heuristic and optimal solution algorithms, as summarized in Campbell, Ernst, and Krishnamoorthy (2002). Some important early contributions include the continuous location hub model of Aykin and Brown (1992), the tabu search heuristic of Skorin-Kapov and Skorin-Kapov (1994) for single allocation problems, and Lagrangean relaxation for models with hub capacities by Aykin (1994). Perhaps the most important advances have come from tighter formulations that allowed larger problems to be solved. Examples include preprocessing methods and new tighter constraints (e.g., Skorin-Kapov, Skorin-Kapov, and O'Kelly 1996; Boland et al. 2004) and application of polyhedral results (e.g., Hamacher et al. 2004; Labbé, Yaman, and Gourdin 2005; Marín, Cánovas, and Landete 2006) to provide much tighter LP bounds.

Much hub location research addressed the fundamental problems or related variants that assumed a constant (flow-independent) discount for transportation on all hub arcs (i.e., α) and required full interconnectivity of the hubs by hub arcs. This may be imposed by the form of the objective or constraints, or implied by the notation, and it often makes sense a priori. These two ideas are related because the economies of scale from concentrating flows on the hub arcs are presumed to create the cost discount for the hub arcs. However, optimal solutions in hub location problems can often result in the hub arcs carrying much smaller flows than do some of the access arcs; yet by assumption the costs are discounted only between the hubs. This clearly represents a limitation of these models because it undermines the basic premise for economies of scale. (This problem can be exacerbated in single allocation models where quite large flows can be concentrated on the single access arc for each node.) Such behavior in solutions is often not apparent because it is only detected though an analysis of the actual flows over the network. The full interconnectivity condition for the hub arcs is also important because it converts the network design problem at the hub level to a location problem for the hubs.

These weaknesses were recognized early on and some research considered different models, such as the hub center model on a tree of Iyer and Ratliff (1990). Chou (1990, p. 245) noted that in future work "the scale factor should be applied to all links … [and] must be a variable depending on the flow volume." A number of researchers have developed models that relax the basic assumption of flow-independent discounts. Hub location models with flow-dependent discounts were developed initially by Bryan (1998) and O'Kelly and Bryan (1998) and were later employed in studies by Racunicam and Wynter (2005), Kimms (2006), and Cunha and Silva (2007). However, most researchers continue to employ a computationally attractive flow-independent discount on all hub arcs.

Other approaches in hub location address the limitations from requiring full interconnectivity of the hubs and assuming a constant discount for transportation on all hub arcs. Jaillet, Song, and Yu (1996) modeled the deployment of two types of aircraft to serve demand, rather than assuming a priori a hub structure. Their model designed the network to handle flows of aircraft and passengers via intermediate cities, though it does not explicitly identify cities as hubs (which is necessarily a rather artificial construct). Chou (1990) employed a similar idea on a tree network where multiple levels of hubs were defined by the volume of traffic they handle in the solution. The hub arc models of Campbell, Ernst, and Krishnamoorthy (2005a) relaxed the restriction of fully interconnected hubs by locating the discounted hub arcs instead. This provides more of a network design perspective by locating the hub arcs (whose endpoints are hubs) that may or may not be connected.

In spite of the recognition of the shortcomings from requiring full interconnectivity of the hubs and assuming a constant discount for all hub arcs, and publication of models that address these shortcomings in several ways, the majority of hub location research has continued to adopt these conditions.

4. Present Status of Hub Location Research

There are several recent surveys and reviews of hub location research (Campbell, Ernst, and Krishnamoorthy 2002; Alumur and Kara 2008; Kara and Taner 2011) and a recent special journal issue (*Computers and Operations Research* 36(12) in 2009). Rather than provide a new review and survey, in this section we seek to highlight some of the most recent interesting research from the past few years—and offer some encouragement for a shift of focus in certain areas. This is necessarily an idiosyncratic presentation of the authors and is not intended as a comprehensive snapshot of all the diverse areas of hub location research. The remainder of this section describes recent works in six areas: solving larger problems, alternative network topologies, integrating cost and service, dynamic models, models with competition, and models with stochastic elements and reliability considerations.

4.1. Solving Large Problems

Much of the published research over the past 25 years has sought to solve larger instances with better mathematical programming formulations and solution algorithms for the fundamental hub location problems or their extensions, especially with fully interconnected hubs and flow-independent discounts. With recent advances, researchers are now able to solve to optimality large hub location problems. Contreras, Cordeau, and Laporte (2011c) solved multiple allocation uncapacitated hub location problems with up to 500 nodes (250,000 OD pairs or commodities) using an enhanced Benders decomposition algorithm. Ilić et al. (2010) developed general variable neighborhood search heuristics to solve large single allocation p-hub median problems and reported results that improved the best known solutions for some 400 node problems and new results for a few problems with up to 1,000 nodes and 20 hubs. Contreras, Díaz, and Fernández (2011) solved large capacitated single allocation hub location problems with up to 200 nodes using a branch-and-price algorithm that integrates lower bounds from Lagrangean relaxation. There has also been impressive progress on hub center problems, as shown in Meyer, Ernst, and Krishnamoorthy (2009), where the single allocation p-hub center problem was solved to optimality with up to 400 nodes using a two-phase algorithm that includes elements of branch-and-bound and ant colony optimization.

Given these recent successes in solving very large scale capacitated and uncapacitated versions of fundamental hub location problems, it is reasonable to ask how much more effort should be directed at solving idealized problems with fully interconnected hubs and flow-independent discounts on hub arcs. Practical transportation hub networks would seem rarely to be larger than those solvable with the current state-of-the-art; furthermore, faster solution methods may be less of a concern for strategic hub network design than for other large operational or tactical combinatorial optimization problems in logistics, such as crew scheduling and vehicle routing.

4.2. Alternative Topologies

A number of researchers have recently developed new models that provide alternatives to having the hubs fully interconnected by discounted hub arcs. The incomplete hub network models of Alumur, Kara, and Karasan (2009) addressed single allocation versions of all four fundamental hub location problems with incomplete, but connected, hub level networks, using CPLEX for solutions. In a similar spirit, Calık et al. (2009) solved incomplete hub covering problems using a tabu search heuristic. Contreras, Fernández, and Marín (2010) studied a problem where hubs must be connected by a tree of hub arcs and solved formulations with a commercial solver. Gelareh and Nickel (2011) provided models for public transport and liner shipping that minimize the cost for both transportation and establishing the hubs and edges, where the hub level is required only to be connected. One final variation on the topology of the network connecting hubs is the use of *isolated* hubs (not connected to a hub arc) along with hub arcs (Campbell 2010). This multiple allocation model does not require the hub level to be connected. Although one might expect that a one hub problem (typically a median) would be fully explored by now, new single hub model formulations continue to provide intriguing formulations issues (O'Kelly 2009; Berman, Drezner, and Wesolowsky 2007).

Several researchers have also explored alternatives to multiple and single allocation for the access arcs. Yaman (2011) considered the multiple allocation p-hub median problem with a limit on the number of access arcs for each node. She also solved several variants of the model, including one that minimizes the fixed cost for access arcs given an upper bound on the total transportation cost and another that minimizes the sum of transportation and access arc costs with flow thresholds. All formulations are solved with CPLEX, though large amounts of memory may be required. Çetiner, Sepil, and Süral (2010) solved a variant of the multiple allocation p-hub median problem where TSP heuristics were used to find routes among the nodes assigned to each hub. An iterative heuristic was used that alternated between locating hubs and designing routes, while adjusting the path travel costs to approximate route costs. This model also includes some constraints as in hub covering models to ensure the hubs were not too far from the nodes.

Given the known shortcomings of a fully interconnected topology for the hub arcs, we encourage continued research on alternative topologies for both the hub level and access level. More generally, we see great opportunities in models of hub location and network design that exploit advances in both areas, especially in telecom network design.

4.3. Integrating Cost and Service

An important limitation of the fundamental hub location models is the bifurcation between the cost-oriented hub median and related models on the one hand and service-oriented hub center and hub covering models on the other hand. For some discussion of hub center and hub covering models, see Alumur and Kara (2008). Although the hub center and hub covering models were originally formulated and proposed as natural extensions of the analogous regular facility center and covering models, the practical motivations for hub center models in transportation seem rather weak. Unlike the strong practical motivations for regular facility center and covering models (e.g., emergency service systems, especially in the public sector), motivations for pure hub center and hub covering models from private sector express delivery systems seem problematic because no costs are included. Applications of hub covering may be more reasonable in telecom, where signal deterioration with some technologies limits the length of arcs in a path. For example, the network design problem with relays (Cabral et al. 2007) and the regenerator location problem (Chen, Ljubic, and Raghavan 2010) include hub covering constraints that limit the maximum length of any arc in a path, but not the length of the entire path. However, without stronger practical motivations from transportation systems, we feel that research on hub center and hub covering models for transport systems without consideration of transportation costs will be largely an (enjoyable) academic exercise rather than an approach to design real-world logistics and transportation systems.

Because both cost and service are important for most transportation hub networks, the value of a hub location model is likely to be enhanced by incorporating both cost and service—and some recent research has moved in this direction. Examples include cost minimizing models with OD travel time constraints as in time-definite transportation (Alumur and Kara 2009 and Campbell 2009) or limits on access arc lengths (Vidović et al. 2011). Lin, Yang, and Chang (2011) also include limits on access arc lengths that represent walking distances in a novel model of hub location for locating bicycle sharing stations (hubs), with costs for walking, riding, bicycle inventories, and constructing bicycle lanes between hubs. Note that service constraints may actually make problems easier to solve by eliminating variables that represent poor service (e.g., long routes). Other alternatives include bi-objective models with cost and travel time as in Costa, Captivo, and Climaco (2008) and penalty-minimizing models (Chen, Campbell, and Thomas 2008). Note that models combining travel cost and travel distance can effectively be considered in the existing framework of the fundamental hub location models by using the weighted average of travel cost and distance in the objective. For example, if β is the relative weight on travel distance, with $(1-\beta)$ the weight for travel cost, then the objective for a p-hub median model could be

$$\begin{aligned}\sum_{i,j,k,m} & W_{ij}\big[(1-\beta)(\chi d_{ik}+\alpha d_{km}+\delta d_{mj}) \\ &\qquad + \beta(d_{ik}+d_{km}+d_{mj})\big]X_{ijkm} \\ &= \sum_{i,j,k,m} W_{ij}(\chi' d_{ik}+\alpha' d_{km}+\delta' d_{mj})X_{ijkm},\end{aligned}$$

where $\chi' = \chi + \beta(1-\chi)$, $\alpha' = \alpha + \beta(1-\alpha)$, and $\delta' = \delta + \beta(1-\delta)$ are the effective transportation rates for collection, transfer, and distribution, respectively. The effective interhub discounts are then a decreasing function of β:

$$\frac{\alpha'}{\chi'} = \frac{\alpha}{\chi} + \frac{\beta(\chi-\alpha)}{\chi^2+\beta\chi(1-\chi)} \quad \text{and} \quad \frac{\alpha'}{\delta'} = \frac{\alpha}{\delta} + \frac{\beta(\delta-\alpha)}{\delta^2+\beta\delta(1-\delta)},$$

which illustrates how an increase in β increases the importance of travel distance and travel time, as with passenger transportation, and attenuates the benefits of consolidation by increasing the interhub discount and diminishing the economies of scale.

Service has also been viewed in terms of congestion rather than travel time (e.g., Marianov and Serra 2003), as in several recent papers. Elhedhli and Wu (2010) included congestion costs in the objective, along with transportation and fixed costs, and developed a heuristic Lagrangean solution algorithm. Koksalan and Soylu (2010) provided a bicriteria model for multiple allocation problems with congestion costs and solve it using an evolutionary algorithm. Camargo, Miranda, and Ferreira (2011) presented a hybrid approach that combines outer-approximation cuts and Bender's decomposition for single allocation hub center problems with congestion modeled in two ways. Finally, Cantazaro et al. (2011) included congestion in a model that integrates hub location, graph partitioning, and routing for a problem motivated by a particular Internet routing protocol.

4.4. Dynamic Hub Location

Nearly all models to date have dealt with hub location problems as a static case—perhaps not surprising given the large one-time expense and sunk capital costs in large hub facilities. However, in some cases hubs are relatively less expensive facilities (also true in telecommunications networks) and their dynamic configurations are certainly open to many applications of dynamic programming or scheduling. In addition, once the hub location problem is solved, day-to-day operational allocations of routes to hubs and modes are clearly amenable to dynamic considerations. Indeed it would be remarkable to find such a stable interaction system that no further adjustments in hub allocation would be needed.

A few hub location research efforts have addressed dynamic problems, including Gelareh and Nickel (2008), which extended earlier work for design of public transport hub networks that does not require a fully interconnected set of hubs to include multiple time periods with costs for operating and closing hub nodes and hub arcs. Contreras, Cordeau, and Laporte (2011a) used branch and bound and Lagrangean relaxation to solve multiperiod uncapacitated multiple allocation hub location problems with fully interconnected hubs. This model includes costs for opening, operating, and closing the hub nodes (but not the hub arcs), where in each time period, new OD pairs may be added and the demand for existing OD pairs may increase or decrease. Campbell (2010) also explored a dynamic problem where isolated hubs can be used in response to an increasing demand in a fixed region and in response to geographic expansion.

4.5. Competitive Hub Location

Although some hub-based transportation systems operate without competition (e.g., public sector systems, such as transit or postal operations), for most private sector hub-based transportation systems there is competition between multiple service providers. When two or more firms compete for customers (e.g., passengers or freight), the optimal hub location, networks, and traffic patterns may differ greatly from those without competition. Thus, competitive hub location problems provide a rich source of realistic and difficult problems. Eiselt and Marianov (2009) extended the pioneering work by Marianov, Serra, and ReVelle (1999) with a more complex function to allocate passengers based on the relative travel time and travel cost provided by the two competitors. A different model of competition is characterized by Stackelberg hub location problems, where the leader firm locates its hubs in anticipation of a later firm (the follower) optimally locating its hubs based on the known locations of the leader's hubs. Sasaki et al. (2009) solved Stackelberg hub arc location problems with an optimal solution approach based on the optimal algorithm in Campbell, Ernst, and Krishnamoorthy (2005b). Gelareh, Nickel, and Pisinger (2010) also model competitive hub location and network design for liner shipping in which a new firm competes with a existing dominant operator on the basis of service time and transportation cost.

4.6. Models with Stochastic Elements and Reliability Considerations

As with other location and logistics problems, incorporation of stochastic issues in hub location is a promising research area that is receiving increasing attention. One example is Sim, Lowe, and Thomas (2009), which provided a single assignment hub covering model where the arc travel time is normally distributed with a given mean and standard deviation. The objective is to locate p hubs to minimize a parameter β while ensuring that the probability is at least a given level that the travel time for any OD pair is at most β. Contreras, Cordeau, and Laporte (2011b) considered a more straightforward variant of the cost minimizing uncapacitated hub location problem to include stochastic demands and transportation costs. Problems with up to 50 nodes were solved using Benders decomposition with the sample average approximation scheme. Yang (2009) addressed uncertainty in demand via a stochastic two-stage model with multiple demand realizations (scenarios). Hult, Jiang, and Ralph (2011) solved the single allocation p-hub center problem with stochastic travel times using cutting planes and Benders decomposition. A different perspective is provided by Lium, Crainic, and Wallace (2009), which links the strategic location perspective and a tactical service network design perspective to show how the hub network structure can arise, not to exploit economies of scale in transportation, but to hedge against uncertainty in demand.

Hub location models with reliability considerations have also been analyzed recently. Reliability issues may arise in several forms in hub networks, including failures of edges and nodes. Kim and O'Kelly (2009) develop single and multiple allocation models to maximize the expected flow transmitted between origins and destinations, where edges and nodes have reliabilities defined as the probability that the hub or edge transmits flows for a given time period without failing. They formulated and solved two new models: the p-hub maximum reliability model and the p-hub mandatory dispersion model. An, Zhang, and Zeng (2011) consider hub failures and design single and multiple allocation networks with backup hubs and alternate routes to maintain service. The objective is to minimize the expected cost of routing all flows using either the primary route when no hub on the route fails or the backup route if a hub on the primary route fails. Other approaches to reliability issues extended from the facility location and network design literatures may well be incorporated, though the resulting problems are likely to be quite difficult because of the hub interactions.

5. Promising Research Directions

The research described in the previous section shows that the field of hub location is now moving in exciting new directions with improved models that better capture important elements of transportation and logistics systems missing from the early fundamental hub location models. Although the different subsections in §4 provide a glimpse of research opportunities, combinations of several of these suggest rich areas for future research. We especially encourage models that incorporate both more realistic transportation costs and service measures along with other relevant aspects.

Although good progress has been made recently in solving large instances of hub location problems with idealized assumptions, more complex and less idealized hub location models provide strong challenges, especially for finding optimal or near-optimal solutions. Although the fundamental hub models serve a useful function, we suspect that future research in hub location may follow a similar path to that in vehicle routing where some research continues on methods for solving very large scale idealized problems (e.g., VRPs or TSPs with tens of thousands of customers), but more research is directed at more realistic problem variants. We posit that a similar approach will be even more useful in hub location because the strategic nature of the problem makes fast CPU times less important in strategic hub location.

Some other promising areas for future hub location research stem from linkages to other areas. One such opportunity in transportation is to better link hub location to network design in general and to tactical transportation service network design in particular (see Crainic 2000; Wieberneit 2008; Contreras and Fernández 2011). This may provide strong benefits from coordinating the longer term strategic hub location decisions with the more routine operational decisions (e.g., scheduling). Likewise, interesting research opportunities will arise from better cross-pollination between transportation and telecom hub location, including better integrating issues such as hop constraints and congestion. In all of these areas, there is a large literature that may be leveraged to develop more insightful hub location models. One final integrative topic that seems ripe for exploration includes the incorporation of environmental assessment models in hub location problems, especially the regarding the implications of flow consolidation.

The fundamental hub location models have been viewed as being broadly applicable across transportation modes and geographies. However, although their abstract nature provides appealing simplicity and generality, it also limits their ability to accurately model the intricacies of any particular logistics and transportation system. Thus, there are many research opportunities of a more applied nature to better model specific air, ground, or water transportation systems and operations, especially regarding the unique features and limitations of transportation modes and application-specific stochastics, congestion, competition, etc.

Hub location research has benefited from the common use of three data sets: the CAB data set (introduced for hub location by O'Kelly 1986a) is for air passenger traffic between 25 U.S. cities from 1970; the AP data set (introduced by Ernst and Krishnamoorthy 1996) is for postal (ground) operations between 200 locations in metropolitan Sydney, Australia; and the Turkey data set (introduced by Tan and Kara 2007) is for ground transportation between 81 cities in Turkey. These data sets provide a valuable common test-bed of real-world instances derived from applications of differing scale and scope. One aspect of the data that merits further attention is the appropriate cost discounting to reflect economies of scale. In the simplest form with a constant discount factor on hub arcs, researchers typically find solutions for a range of α values (e.g., $\alpha = 0$, 0.2, 0.4, 0.6, 0.8, 1.0). However, in any particular application, the range of appropriate values for the discount factor is likely to be limited, for example, because of the transportation technologies deployed. For example, values range from 0.1 for high-speed rail (Blanco, Puerto, and Ramos 2011) to 1.0 for LTL trucking (Cunha and Silva 2007). Thus, more research would be useful to document the appropriate transportation cost discounts in applied settings. Special consideration is needed for telecom, because hub arcs may have a cost structure similar to that of access arcs because the transmission media (e.g., fiber) is often the same.

The hub location research community has grown quickly in the past quarter-century, and distinctive subgroups have formed that tackle particular aspects of hub problems. Research on strong mathematical formulations, including effective cuts and strong lower bounds, has produced dramatic increases in computational speed and delivered new insights. Such papers are often illustrated with relative computation times, and they rarely they delve into the substantive interpretation of the transportation and logistical results. For example, many papers do not show actual physical location and allocation results or document flows. In contrast, many papers with more of a geography and regional science perspective quite naturally focus on the locational (spatial) aspects, including hub locations, assignments of non-hubs, traffic flows, and their properties. Again, it is a matter of training and preference as to which style of analysis any individual investigator devises, and both aspects are vital and have contributed to the impressive progress in the field. Generally, from our perspective, a focus on algorithms and CPU times to the exclusion of hub location and hub network design is to miss an important piece of the puzzle. To paraphrase Geoffrion (1976, P. 81), "The purpose of hub location research is insight, not numbers"—and better insight is likely to result from the combination of improved understanding of the properties of hub location problems, better algorithms, new solutions, and a clearer understanding of the practical implications of the solutions.

6. Conclusion

Much of the today's hub location research was motivated by O'Kelly's seminal publications from 25 years ago, though the roots of hub location stretch further back to early works in location analysis and network design. Hub location problems involve the location of hub facilities and routing of flows from origins to destinations via the hubs. Hub networks are very common in large-scale transportation and telecom systems, and by consolidating/concentrating flows they help to reduce costs, increase service frequencies, and hedge against uncertainty. Transportation hub location problems involve locating hub facilities to handle flows of freight or passengers, whereas telecom hub networks provide for the movement of electronic messages. We have presented a broad definition of hub location problems, described the relationship between hub location and network design problems, and discussed important motivations for the emergence of research on hub location problems. We described the early research on hub location problems, highlighted some key methodological issues, and gave an overview of some interesting areas of current hub location research. We have also sought to call attention to some shortcomings of the research and provide encouragement for new research to address these shortcomings.

From modest beginnings, hub location, especially in transportation, has grown to become a prominent area of location research with hundreds of published articles—and recent surveys of location science place hub location alongside other major topics in the field. The blend of location and network design aspects gives rise to special challenges for hub location, and the vibrant growth of the past 25 years has provided a solid and strong foundation. Although the motivations for future hub location research may be different from those that stimulated the birth of the field, we foresee a continuing need for good hub location research and promising future for the next quarter-century and beyond.

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